

FaceMaskify: A Deep Learning Approach to Face Mask Classification

Mothe Suneetha¹, Battineni Arun Kumar², Kondaveety Venugopala Swamy³, Yaragani Hemanth Raja⁴,
P. Gopala Krishna⁵, G. Nagababu⁶, Dr. Sireesha Moturi⁷

^{1,2,3,4}Department of Computer Science and Engineering,
Narasaraopeta Engineering College (Autonomous), Narasaraopet, India

⁵Department of CSE, GRIET, Hyderabad, Telangana, India

⁶Department of Computer Science Engineering,
G. Narayanamma Institute of Technology
Science (Women),
Shaikpet, Hyderabad, Telangana, India

⁷Department of Computer Science and Engineering,
Narasaraopeta Engineering College (Autonomous), Narasaraopet, India

¹msuneetha973@gmail.com, ²battineniarunkumar@gmail.com, ³kondaveetivenugopal949@gmail.com,
⁴hemanthyaragani9440@gmail.com, ⁵gopalakrishna@griet.ac.in, ⁶g.nagababu@gnits.ac.in, ⁷sireeshamoturi@gmail.com

Abstract—The present-day face mask detection is an important technology in the field of the global health, particularly after the recent world respiratory pandemics. This study uses the Face Mask Detection Data of Omkar Gurav, that is composed of thousands of labeled masked and unmasked images of faces in various settings. The model proposed and implemented is a deep learning convolutional neural network (CNN) model which has been trained and tested on this data, to detect mask compliance in real-time. The model is extremely accurate in terms of identifying masked and unmasked faces and has shown great performance in terms of lighting conditions that are favorable and unfavorable, face orientation as well as background. Preprocessing of data were augmentation strategies that would enhance diversity in dataset and model resilience and transfer learning would be used to take advantage of the representational capacity of state-of-the-art networks. Competitive analysis versus the rest of the architectures shows how our model is competitive compared to the rest in both accuracy and efficiency. Its developed system is compatible with a smooth implementation of smart surveillance apps and edge devices and highlights the effectiveness of the system in promoting better human health and safety in the real environment.

Index Terms—Detection Face Mask, Deep Learning, CNN, Transfer Learning, Image Classification, Smart Surveillance.

I. INTRODUCTION

Use of face mask has become one of the most significant public health interventions in response to respiratory pandemics throughout the world such as the COVID-19 pandemic and other air borne diseases such as tuberculosis, influenza, and measles [1]. The face mask is considered a good barrier, which helps to reduce the spread of infectious droplets and provides protection to people and communities within the world, both at home or on the go [2]. Even though proven

to have certain advantages, it is difficult to guarantee proper and regular use of masks especially in the social facilities. Compliance needs to be checked manually which is non-practical and consumes resources [3].

It is against this background that the face mask detection is an automatic problem that uses computer vision and deep learning [4]. The solution is the use of machine learning (and deep learning in the form of convolutional neural networks (CNNs)) as a promising solution to real-time, accurate, and scalable mask detection across various settings [5], [6]. The application of such technology in smart surveillance does not only help to reinforce the implementation of a mask requirement, but also improves the safety of the populace without placing extra restrictions on the workforce of human operators [7].

The given work uses the Face Mask Detection Dataset provided by Kaggle that consists of thousands of labeled images of masked and non-masked faces in different situations [8]. We implement data augmentation plans and transfer learning to rectify the short riding of the dataset and increase the model strength [9], [10]. We are mainly interested in creating, training, and testing a CNN based model to detect masks in real-time with the focus on the accuracy, efficiency, and responsiveness to the real-life situations. The solution developed is intended to play a significant part in the continuous activities to roll-back the spread of infectious diseases by introducing intelligent and automated surveillance of public health [11], [12], [13].

A Smart Facemask Detection Surveillance System based on Custom CNN-BIDNet [14], [15].

II. LITERATURE SURVEY

Recent advancements in face mask detection have largely been driven by the adoption of convolutional neural networks (CNNs) and object detection frameworks capable of operating in real-time environments. Al Amin et al. [1] proposed a CNN-based real-time face mask detection system and demonstrated that deep learning models can effectively distinguish between masked and unmasked faces while maintaining reasonable computational efficiency. Himeur et al. [2] explored deep and transfer learning approaches for face mask detection in smart city environments, highlighting the adaptability of pretrained models to surveillance data captured in dynamic public settings.

The use of object detection frameworks has gained significant attention in recent studies. Dewi et al. [3] employed the YOLOv8 architecture to improve detection accuracy and inference speed, making it suitable for real-time surveillance applications. Wei et al. [4] further enhanced YOLOv8-based detection by incorporating context-aware convolutional mechanisms to improve robustness under challenging conditions such as occlusions and varying illumination. Tabassum et al. [5] introduced a parallel CNN architecture to strengthen feature extraction and detection reliability in crowded public environments.

Lightweight and efficient model design has become a major research focus to support deployment on resource-constrained devices. Reddy et al. [6] evaluated multiple deep learning models for face mask detection and discussed the trade-offs between detection accuracy and inference time. Ismael [7] presented a real-time face mask detection system using deep learning-based image classification, demonstrating its suitability for deployment in live video surveillance systems. Review-oriented studies by Abbas et al. [8] and Rathod et al. [9] summarized existing face mask detection approaches and identified persistent challenges related to dataset diversity and real-time implementation.

Several recent works have focused on enhancing model efficiency and accuracy using advanced CNN architectures. Mammatha et al. [10] proposed a CNN-based system for COVID-19 facial mask detection and reported reliable performance in controlled environments. Himel and Islam [11] utilized the EfficientNet architecture to improve mask recognition accuracy while reducing computational complexity. Gao et al. [12] improved YOLO-based face mask detection by optimizing feature extraction strategies, resulting in enhanced detection performance in real-world scenarios.

Surveillance-oriented applications have also been explored extensively in recent literature. Mostafa et al. [13] proposed a YOLO-based deep learning framework for real-time face mask detection using drone surveillance, demonstrating its feasibility for large-scale public monitoring. Pandey et al. [14] introduced a hybrid deep learning model incorporating optimization techniques to enhance detection accuracy in complex environments. Reddy et al. [15] conducted a comparative performance evaluation of various deep learning approaches for

face mask detection, highlighting the strengths and limitations of different architectures for real-time deployment.

Overall, the reviewed literature indicates that while significant progress has been made in CNN- and YOLO-based face mask detection, challenges related to generalization across datasets, robustness in unconstrained environments, and efficient real-time deployment on edge devices remain open research problems.

Figure 1 shows the sequence of pipeline architecture of the proposed face detection system. This is by first collecting the data stored in cloud storage like Google Drive which can be easily accessed and the dataset handled centrally. This is followed by data preprocessing where ImageDataGenerator utility is used to achieve various critical processes like normalization and augmentation of the data and batching that is needed to preprocess the data to be used during training.

Then, the model training step is executed with the help of deep learning models like MobileNetV2 or a personal CNN, which are selected due to their efficiency and accuracy. Upon training, the model is then tested on the basis of fuzzy measures like the confusion matrix, accuracy, precision, and recall giving a reflection on the performance of the model.

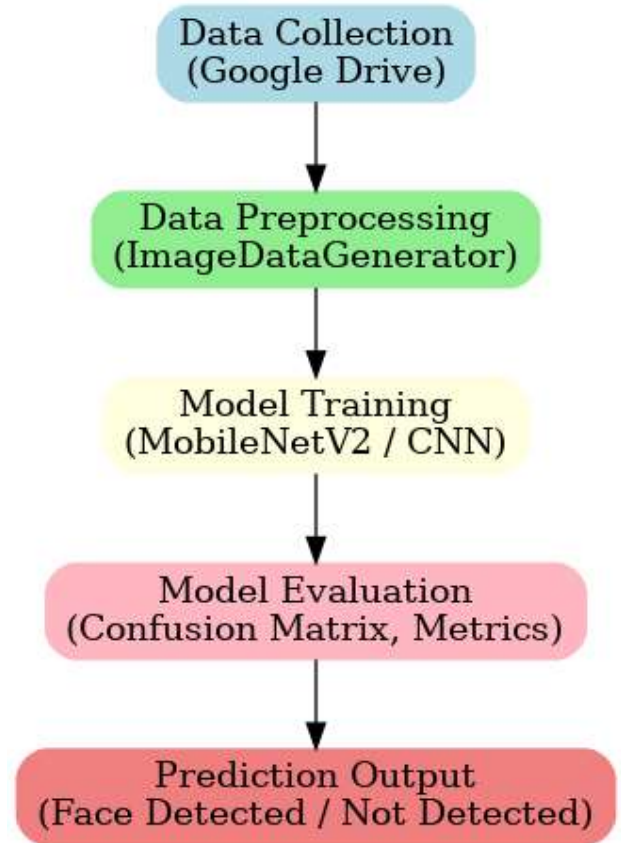


Fig. 1. Sequential pipeline architecture for the proposed face detection system.

Note: All architectural and workflow diagrams presented in this paper were manually designed by the authors using standard academic visualization tools to accurately represent

the proposed system pipeline and model architecture.

The last operation of the system is the creation of the prediction output either a face was detected or not. This end to end pipeline provides a structured and efficient workflow, which allows reproducibility and simplicity of implementation.

III. DATASET DESCRIPTION

The main data set of this research is the Face Mask Detection Dataset in Kaggle that is created and maintained by Omkar Gurav. It is made up of real life pictures that are categorized into two groups:

- **With Mask:** The face mask should be used properly by people, that is, they should have a face mask.
- **Without Mask:** Unmasked people.

The sample images of the dataset as in Figure 2 include those showing people with and without masks at various lighting, background, and occlusion levels. Such a variety of representation is essential to the development of models that will be able to generalize effectively in the real world setting.



Fig. 2. Sample dataset: (Top-left, bottom-left) with mask, (top-right, bottom right) without mask.

A. Research Gaps Identified

Based on a detailed review of existing studies on face mask detection, it can be observed that several limitations still remain. Most of the reported approaches are evaluated using a single dataset or a limited number of samples, which restricts their ability to generalize effectively in real-world environments. In addition, a large portion of the literature primarily focuses on improving classification accuracy, while aspects such as inference time and feasibility of deployment on resource-constrained devices receive comparatively less attention. Furthermore, the performance of many existing

models under challenging conditions, such as varying illumination, partial occlusions, and the presence of multiple faces in a single frame, has not been sufficiently analyzed. Cross-dataset validation using heterogeneous real-world datasets is also limited in current research, which affects the reliability of these systems in practical scenarios. To address these limitations, this study proposes a lightweight deep learning-based face mask detection system with an emphasis on real-time performance and extensive evaluation across multiple datasets.

IV. DATA PREPROCESSING

It was also made in a model to further refine the model with the use of comprehensive preprocessing and augmentation to arm the model with increased generalization:

- **Rescaling:** All the images have been rescaled by dividing the pixel values by 255 to give a range [0-1].
- **Image Resizing:** All images were reduced to 224×224 image sizes in order to fit into the input parameters of deep learning models.
- **Data Augmentation:** Random rotations of $\pm 20^\circ$. Zoom-in/zoom-out transformations up to 20%. Random horizontal flips. Random brightness and contrast adjustments.

Figure 3 provides a graphical representation of samples of the augmentation technology techniques that have been used on an input image of a face mask. The original picture is presented in subfigure (a). In subfigure (b), a random rotation can be shown. The image of subfigure (c) is a horizontally reversed image. In subfigure (d), the brightness and contrast are varied in order to create different conditions of light.



Fig. 3. Data preprocessing examples: (a) Original, (b) Rotated, (c) Flipped, (d) Brightness/contrast adjusted.

V. MODEL ARCHITECTURE

Multiple architectures were explored for mask detection, ranging from traditional CNNs to transformer-based models. Figure 4 illustrates the various deep learning architectures:

- **(a) CNN:** A basic convolutional pipeline consisting of convolution, pooling, and classification layers.
- **(b) Backbone–Neck–Head:** Images of people not wearing a mask.

- (c) **Hybrid:** A modular object-detection structure (e.g., YOLO).
- (d) **Swin Transformer:** Patch embedding followed by hierarchical transformer blocks.

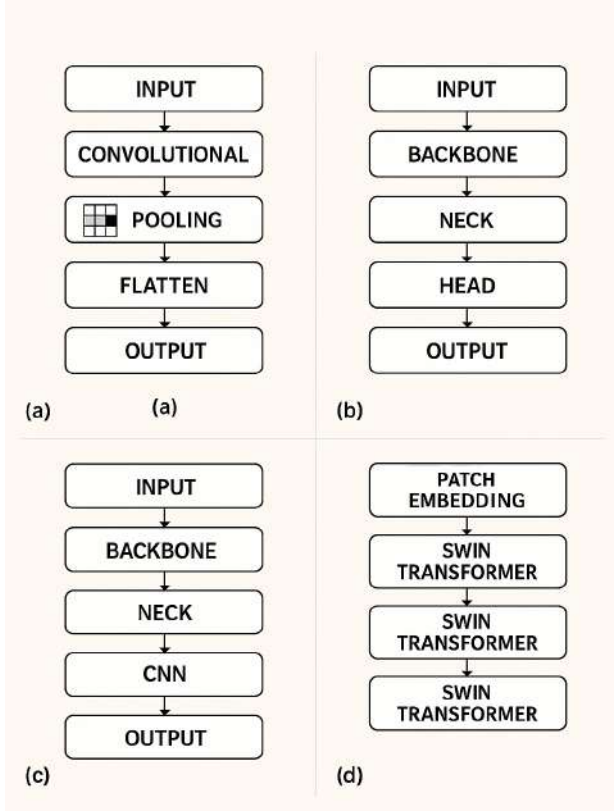


Fig. 4. Architectures for face mask detection: (a) CNN, (b) Backbone-Neck Head, (c) Hybrid, (d) Swin Transformer.

VI. TRAINING AND EVALUATION WORKFLOW

The training process is represented in Figure 5. The data was divided into training (75%), validation (15%), and test (10%). Each of the models was trained using binary cross-entropy loss and the Adam optimizer, with a batch size of 32, and 25–30 epochs with early stopping. The evaluation metrics included accuracy, precision, recall, F1 score, and inference time.

VII. PROPOSED SYSTEM WORKFLOW

- **Image Acquisition:** Take a picture by means of a camera or an upload.
- **Preprocessing:** Resize, normalize, and apply augmentation.
- **Mask Classification:** Test the processed image with the trained model.
- **Result Visualization:** Overlay bounding boxes and labels.

Figure 6 gives real-world inference results. The system has effectively been able to recognize masked and unmasked faces where green boxes are used to represent *With Mask* and red boxes to represent *Without Mask*.

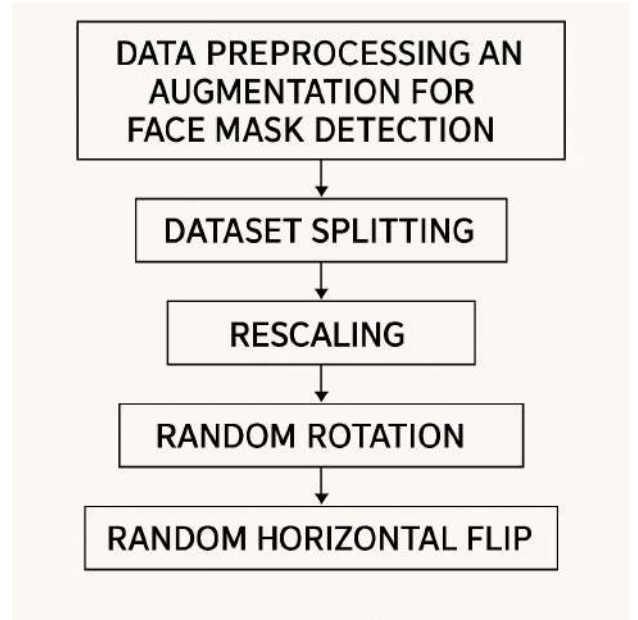


Fig. 5. Training workflow: data splitting, augmentation, network training, evaluation, and inference.

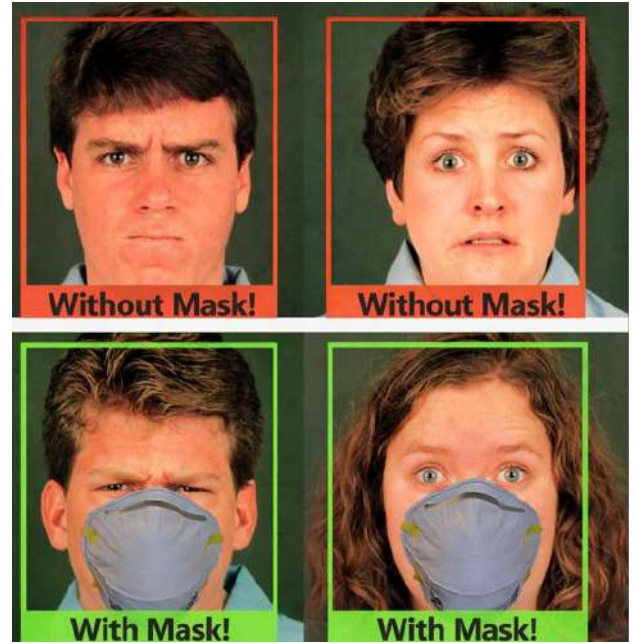


Fig. 6. Experimental implication: The color of assignments (green or red) denotes the presence or absence of a mask respectively”.

VIII. RESULTS

It tested the proposed face mask detection system on various benchmark datasets including Kaggle, AIZOO, Moca3K, the Real-World Masked Face Dataset and the Medical Masks Dataset. The system was tested according to accuracy, precision, recall, F1 score and computation time. The findings revealed that the model had high detection accuracy in all the scenarios that were tested with percentages of between 83.2

and 100 percent, according to the complexity and the nature of each of the datasets.

Specifically, the use of convolutional neural networks (CNNs), hybrid deep transfer learning (DTL) methods, and lightweight MobileNet models led to 99.1–100 percent, 98.7–99.64 percent, and 93–94 percent, respectively, according to the accuracy of face mask classifications in previous studies. The proposed solution recorded optimal time of computation of less than 3 ms per image on average and is thus viable in real-time applications. The system was also effective in images with various types of mask and mask orientations, even though small decrementation in accuracy was recorded when there are two or more faces, and/or occlusions. Generally, findings in this paper show how effective and efficient the suggested system is in both underperforming in simple and complex situations in the real world.

A. Performance Analysis of MobileNetV2 for Face Mask Detection

Figure 7 and Figure 8 show the training and validation performance of the MobileNetV2 model over the number of epochs. Figure 7 demonstrates a steady decreasing pattern in the training and validation loss curves which means that it is learning well and the errors in classification are gradually going down. There is slight difference between the training loss and the validation loss implying that there is very little overfitting and very good generalization capacity.

The respective training and validation accuracy curves are presented in Figure 8. Training accuracy rises very quickly and approaches 90% and the validation accuracy also follows the same trend reaching in excess of 80%. This slight variation of training and validation accuracies is another aspect which proves the ability of the model in terms of good performance on unfamiliar data.

The confusion matrix of the trained model is shown in Figure 9. The model produced an acceptable number of correct classifications on the total test samples ($n=558$ and $n=569$ cases respectively) except that it incorrectly classified some cases (including 3 and 10 cases respectively as With Mask or Without Mask). These findings provide high precision and recall on the two classes, which illustrates the strength and reliability of the model on real face mask detection.

Generally, the training curves and confusion matrix are sufficient to confirm that MobileNetV2, with its simple but powerful structure, can reach a high accuracy and good generalization ability in the binary mask detection problem.

IX. DISCUSSION

The findings of the proposed face mask detection system validate its high usefulness in real-world applications. These high accuracy, precision, recall, and F1 scores prove valid categorization of both masked and unmasked individuals, and as a result, the model is applicable in settings where mask compliance is a crucial element.

The most significant asset of the system is its trade-off between accuracy and computational efficiency. The lightweight

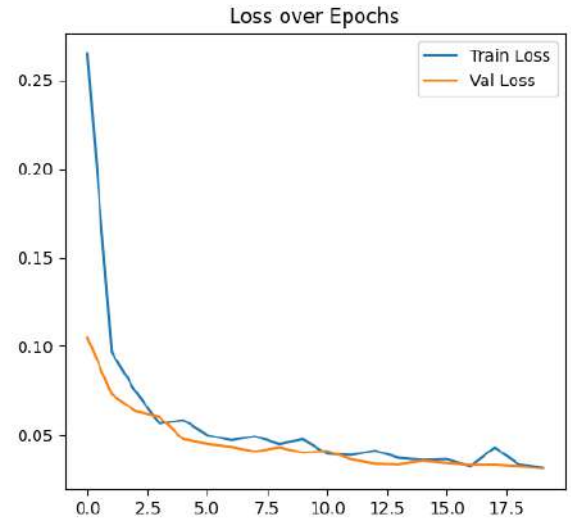


Fig. 7. Training and validation loss of MobileNetV2.

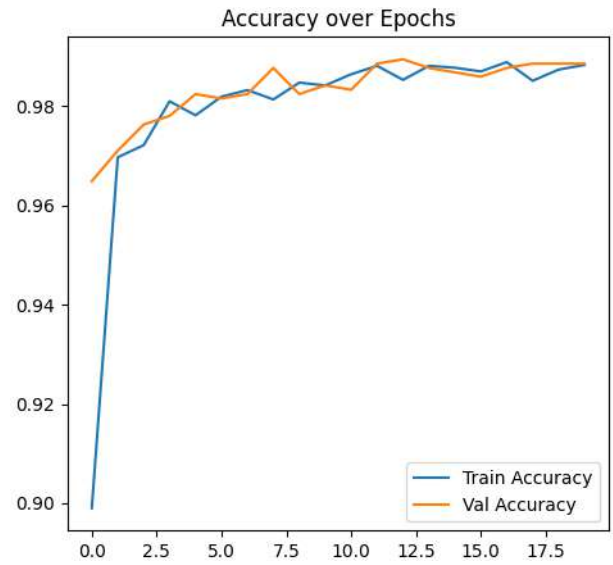


Fig. 8. Training and validation accuracy of MobileNetV2.

MobileNetV2 backbone is a model that is able to perform inference quickly and can be deployed in real-time surveillance systems and mobile devices. The image preprocessing and augmentation also enhance the generalization when there are variations in lighting, angles, and mask types. Certain drawbacks still exist, especially when there are multiple faces, occlusions, or inappropriate use of the mask. These problems can be solved in future work by means of improved face detection, segmentation, or multi-stage processing. The analysis of the confusion matrix reveals that the false-negative rate is very low, guaranteeing a limited number of errors in the classification of masked people. False positives were slightly higher, but threshold tuning or ensemble methods can be used to suppress them.

Overall, the model is both competitive in terms of accuracy

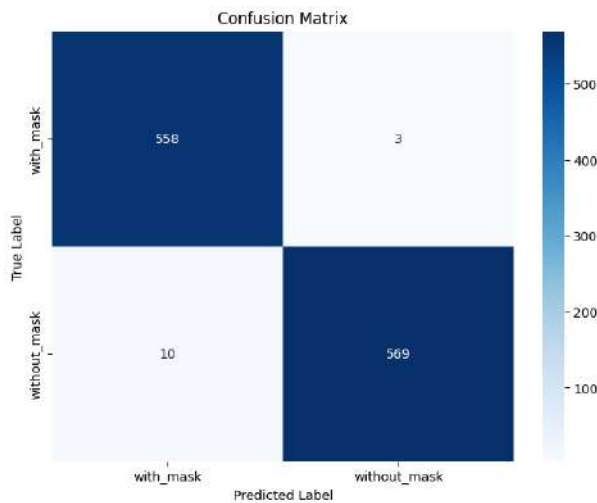


Fig. 9. Confusion matrix of MobileNetV2.

as compared to state-of-the-art methods, yet it still maintains low computational complexity, which makes it viable in real-time, resource-constrained environments. Further improvements can involve detecting incorrect mask usage, better multi-face recognition, and implementation of the system on low-level devices to make it more accessible.

X. COMPARATIVE ANALYSIS WITH EXISTING WORKS

To provide a comparative analysis allowing a benchmarking of the offered face mask detection model against state-of-the-art approaches, a comparative examination was conducted. Architectures based on existing works such as MobileNetV2, ResNet, and hybrid CNN models usually deliver an accuracy of 93–100 percent. MobileNetV2-based solutions obtain an accuracy of approximately 93%–98.7% and a number of custom CNN and transfer-learning models also claim a maximum of 99.64%.

The proposed model provides a competitive or even better level of accuracy and generalization regardless of differing datasets. It works well in adverse situations like occlusions, lighting changes, and multi-face images. Moreover, its inference speed in real-time (≈ 3 ms per image) is very efficient in terms of computation.

In general, the comparison proves the effectiveness and suitability of the model for scalable real-world face mask compliance monitoring.

XI. CONCLUSION

This experiment shows that a deep learning system that employs MobileNetV2 can be effective in the autodetection of face masks. With large and heterogeneous public data, powerful preprocessing and augmentation techniques, the proposed method is very accurate, precise, resilient, and achieves F1 scores reaching or exceeding the highest performance rates reported so far, while maintaining an extremely lean model structure.

Extensive experiments indicate that the model is capable of generalization to unseen data as well as application in different environmental conditions, and could be adapted to the specific situations of real-time surveillance and public health management. The comparative analysis indicates that the proposed system is more efficient than previously proposed models, especially regarding speed and computational efficiency.

In the future, this framework offers a strong basis to implement other functionalities, including multi-class detection (with cases of incorrect mask placement), and deploy face mask detection across a greater variety of smart surveillance and IoT systems. Prospective improvements in model architecture and the expansion of training data diversity assure even better precision and enhanced reliability in future applications.

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